Hamid Mahmoudi PLC-HW01
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1,

فرض:
اعدادی که با 0 شروع بشن و بیشتر از 1 رقم باشن int نیستن

int = ([1-9])([0-9])* + 0

pos_int = ([1-9]) ([0-9])* ('n' + "pos")

u_int = (0 + ([1-9])([0-9])*)('u' + 'w')

neg_int = ?

- کلا زیر مجموعه int است و چون پسوند/پیشوند ندارد نیازی به ساختن توکن جدید برایش نیست

float = ((int)('.')(int) + (int)('.') + ('.')(int))('f' + ε)

int منظور همان RE بخش اول است.

double = ((int)('.')(int) + (int)('.') + ('.')(int))('d')

sci = (int + float)('e')(int)

برای بخش exp بجای int می‌شود ('+'+'-'+ε)(int) گذاشت
ولی باتوجه به توضیحات پایانی منظور اصلی را متوجه نشدم.

cbase_int = ([1-9])([0-9])* ('$')(int)

bin = ('0')('b'+'B')('0'+'1')*

quad = ('0')('q'+'Q') ([0-3])*

فرض:
0 به صورت 000 و ... نوشته نمی‌شود و به همان شکل 0 ورودی می‌شود
[illegible]

octal = ("0c")([0-7])*

decimal = ((int)('.'+ε)(int+ε) + ('.')(int))('m')

hex = ('0')('X'+'x')(H)* + [0-9](H)*

HEX = [0-9]+[a-f]+[A-F]

dhex = ('0')('X'+'x') ((H)$^+$('.')(H)* + ('.')(H)$^+$) + ('.')(H)$^+$ + ([0-9])(H)*('.')(H)*
, fhex

scihex = (hex + dhex)('p')(int)

az_int = ('0'+('x'+'A'))([0-9]+[a-z]+[A-Z])$^+$

2)

اولویت	Rule	RE
1	A	ab^*c
3	D	a^*c
4	E	$a^*b^*c^*$
2	R	$(a+\varepsilon)bc^*$

$$\underbrace{bcc}_{\substack{E\\ or\\ \boxed{R}\\ \downarrow\\ R>E}}\ \underbrace{abbbcc}_{\boxed{E}}\ \underbrace{abc}_{\substack{\boxed{A}\\ or\\ E\\ or\\ R\\ \downarrow\\ A>R,E}}\ \underbrace{aac}_{\substack{\boxed{D}\\ or\\ E\\ \downarrow\\ D>E}}\ \underbrace{aabbc}_{E}\ \underbrace{bc}_{\substack{E\\ or\\ R\\ \downarrow\\ R>E}}$$

$\Rightarrow A > R, E \qquad R, D > E$

D می‌تواند ۱,۲,۳ باشد و A , R به صورت
یکتا در هر حالت مشخص می‌شوند.

$$\underbrace{aaac}_{D}\ \underbrace{abc}_{A}\ \underbrace{bcc}_{R}\ \underbrace{aabcc}_{E}$$

---

3)

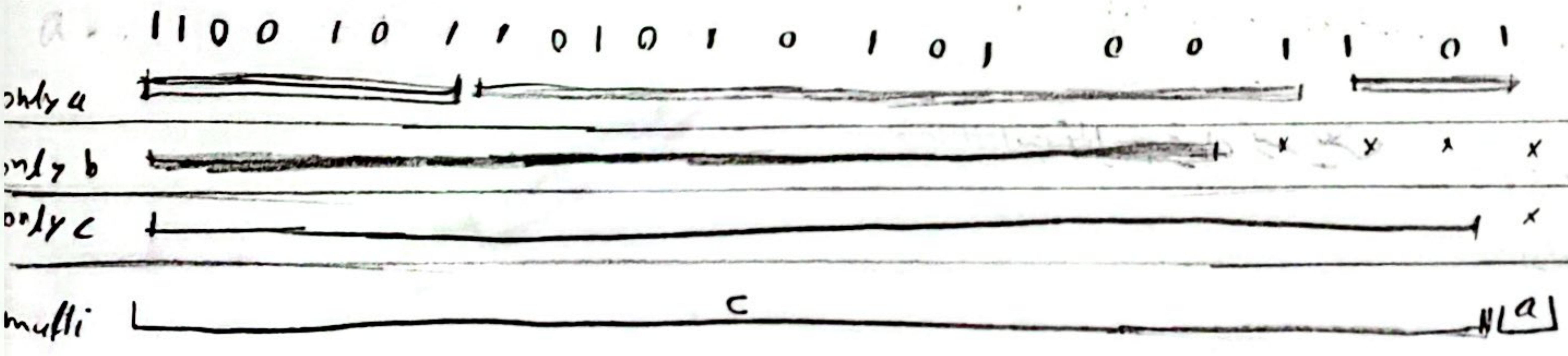
الف)

a. $(1^*)(0^+(1+\varepsilon))^*$

b. $(0+1)^*(00)$

c. $(11)(0+1)^*(0)$

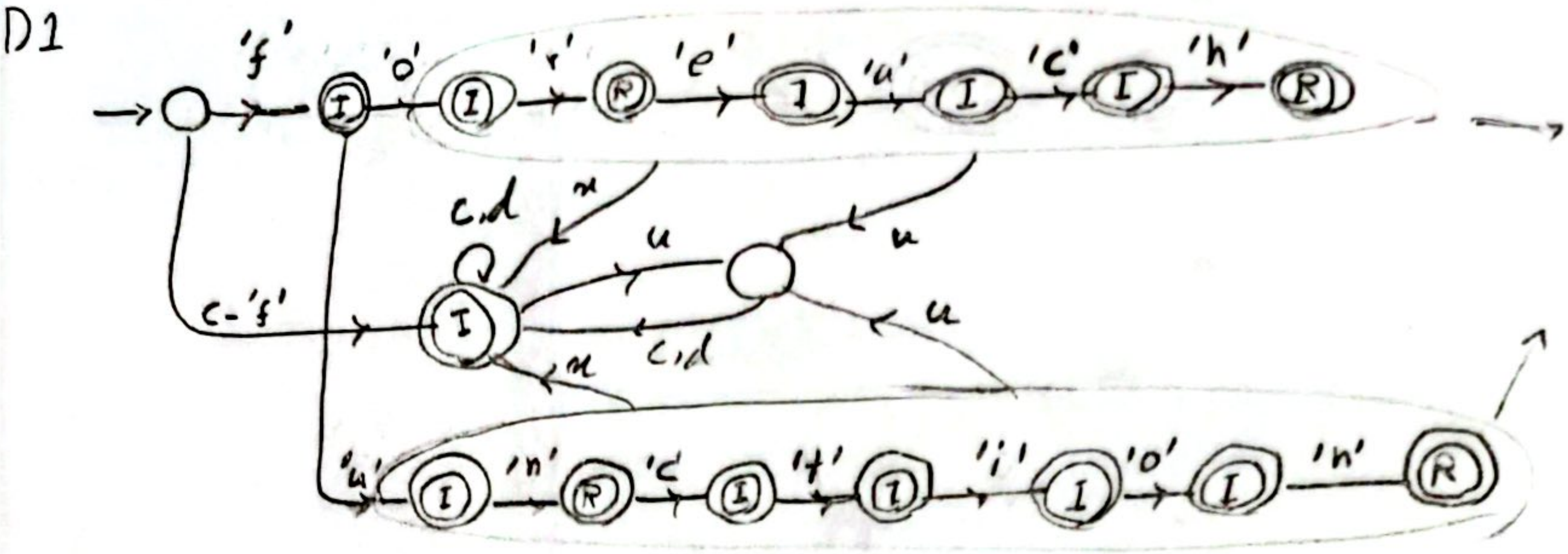
ب)



4،

الف، accepting states : I → ID , R → RES

D1

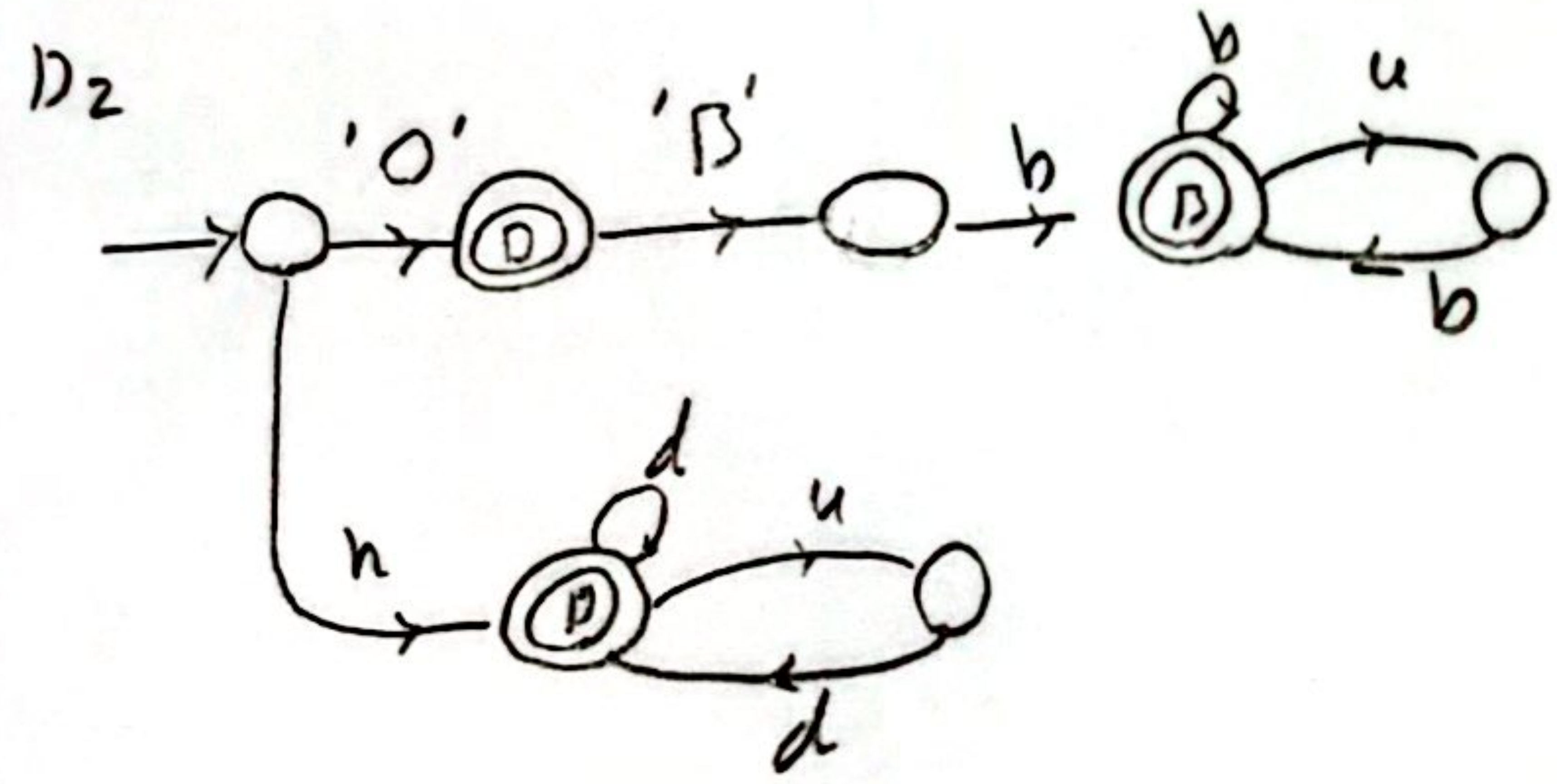


x برای این مسیرها می‌شود
c,d - previous
یعنی یا مثلی که
یک عدد char
است یا لاتین واردش
شده است.

بقیه حالات به Trap بروند

ب، accepting states : B → BIN, D → DEC

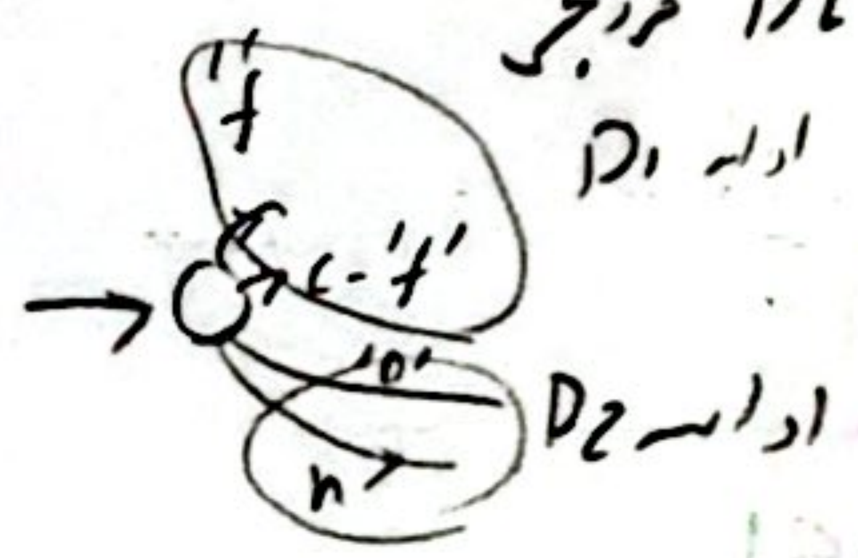
D2



0 → D
00 → X
12_34 → D
-12 → X

بقیه حالات به Trap بروند

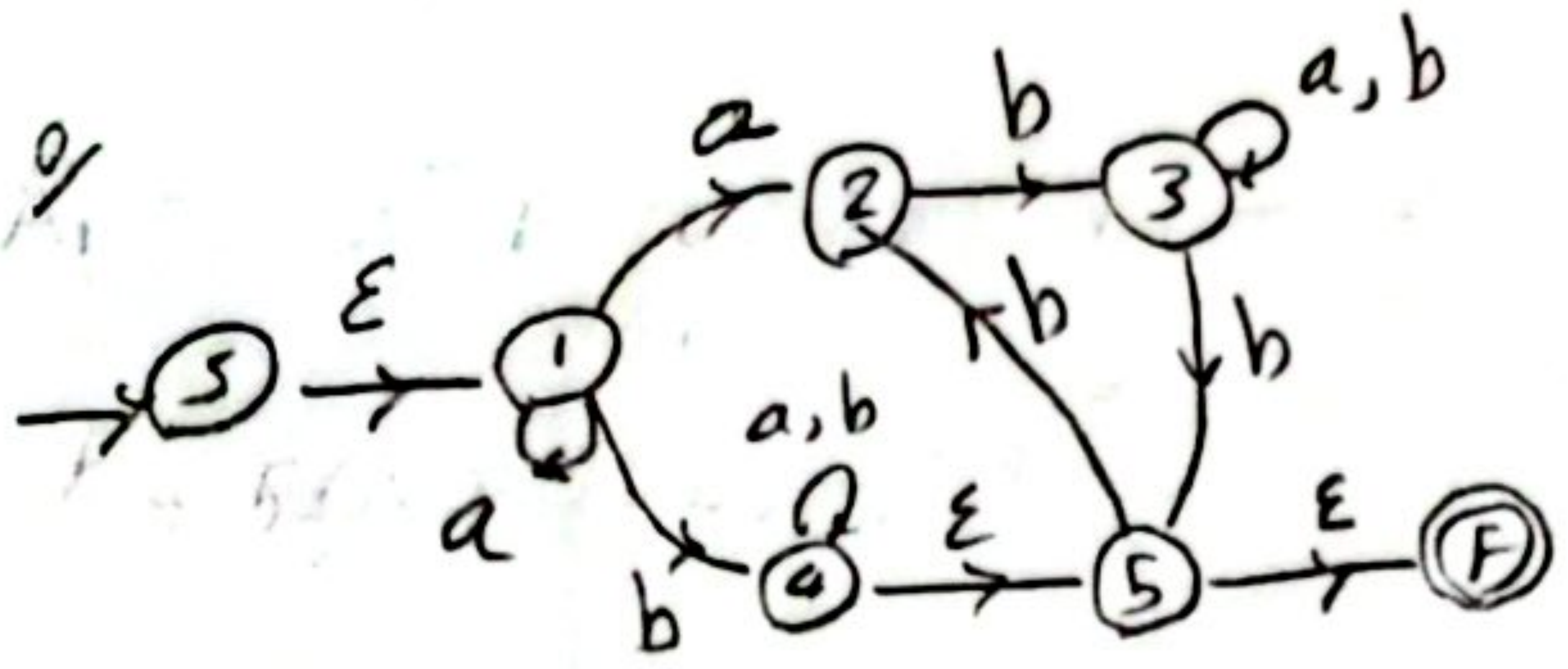
پ، چون شامل ID, RES هندل شده و از قبل بیشترین انطباق نیز در طراحی D1 و D2 هست و همچنین استیت اولیه D1 و D2 خروجی یکسان ندارد کافیست هر دو از استیت اولیه منتقل کنیم یعنی دیگر خروجی‌های اولیه نیز از Trap خارج می‌شوند.



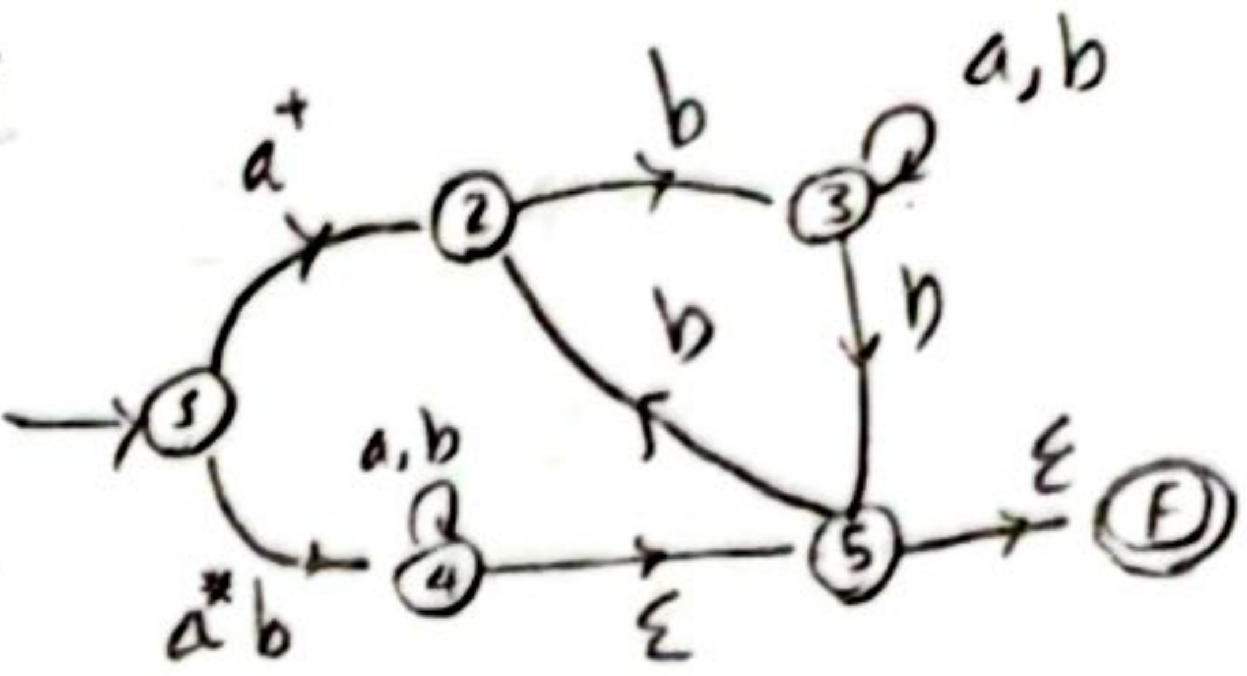
ت، چراکه تحلیل‌گر لغوی بلندترین رشته‌ای که حداقل با یکی از Token ها صدق کند را می‌یابد و سپس اگر چند توکن داشت اولویت بندی می‌کند؛ در این سوال نیز چون function2 طولانی‌تر از function است باید ابتدا کل آن را Token در نظر بگیرید.

5)

الف) 0/



1/



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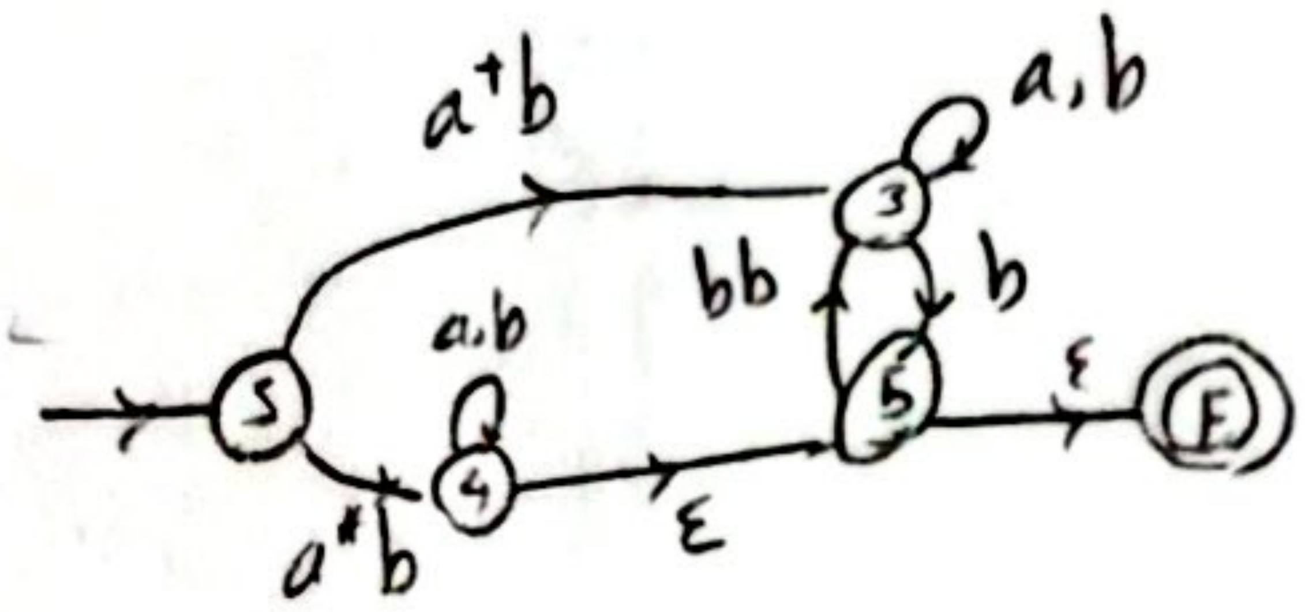
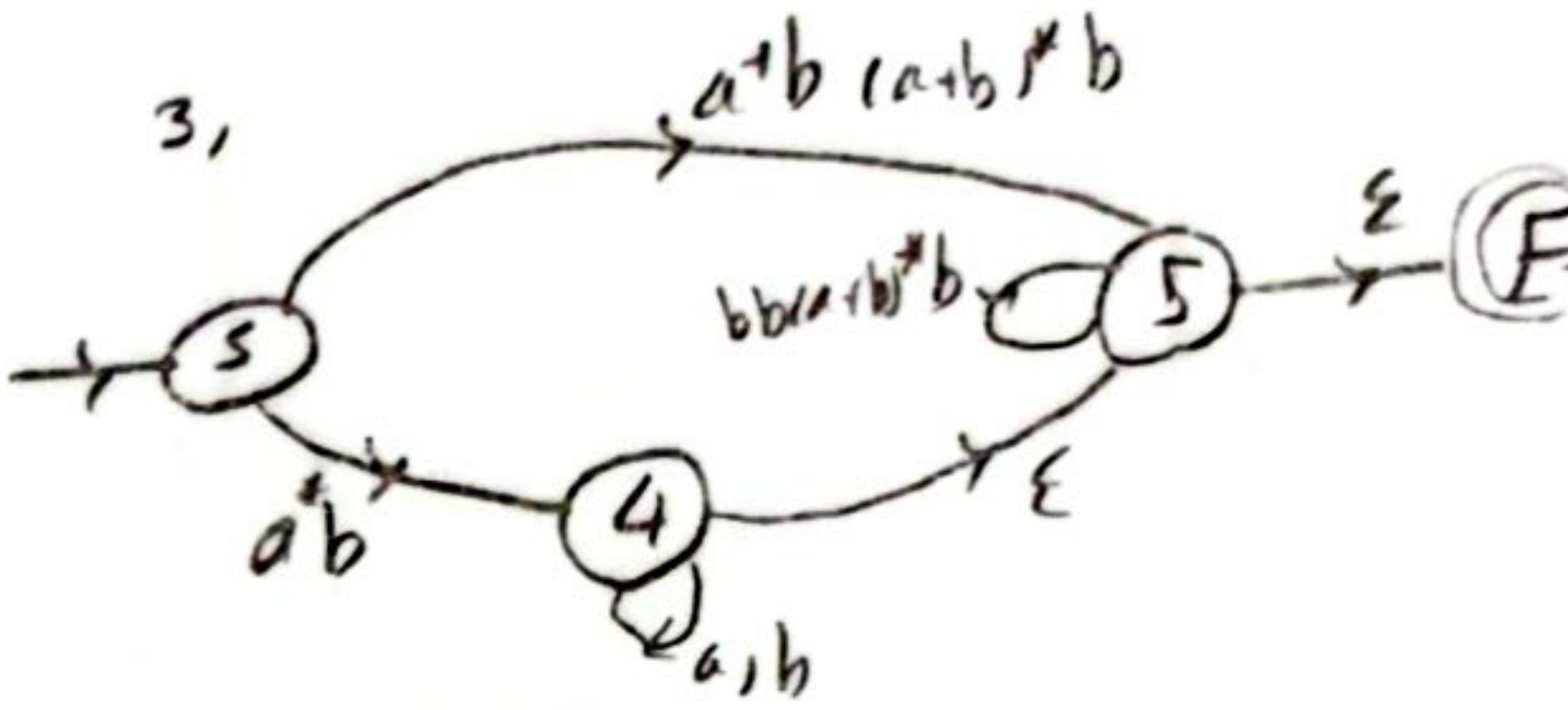
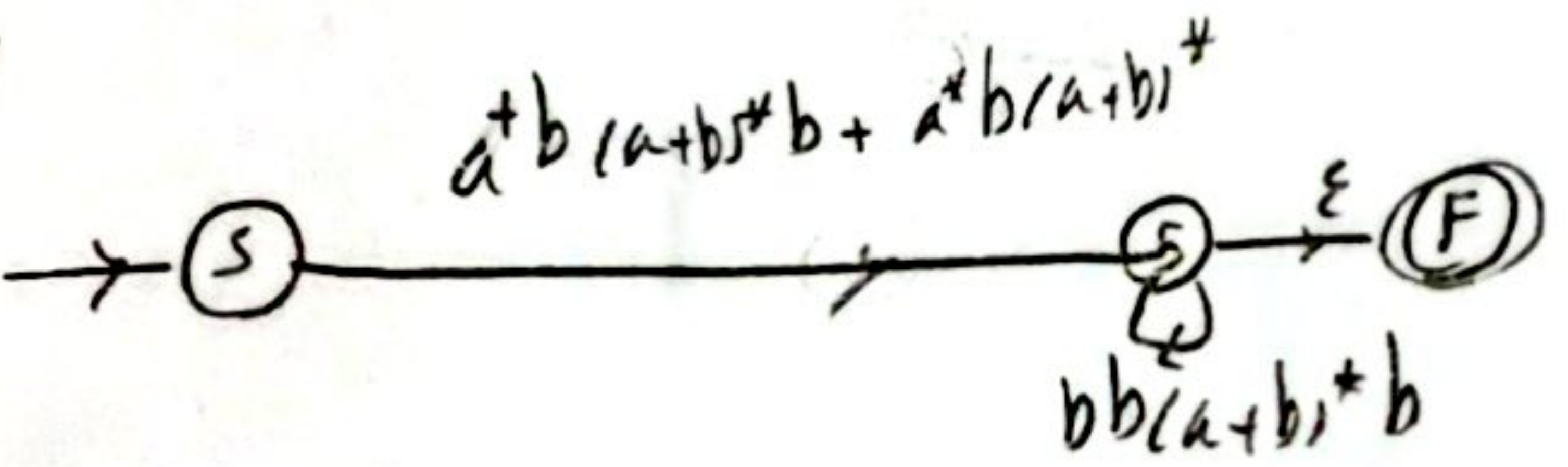


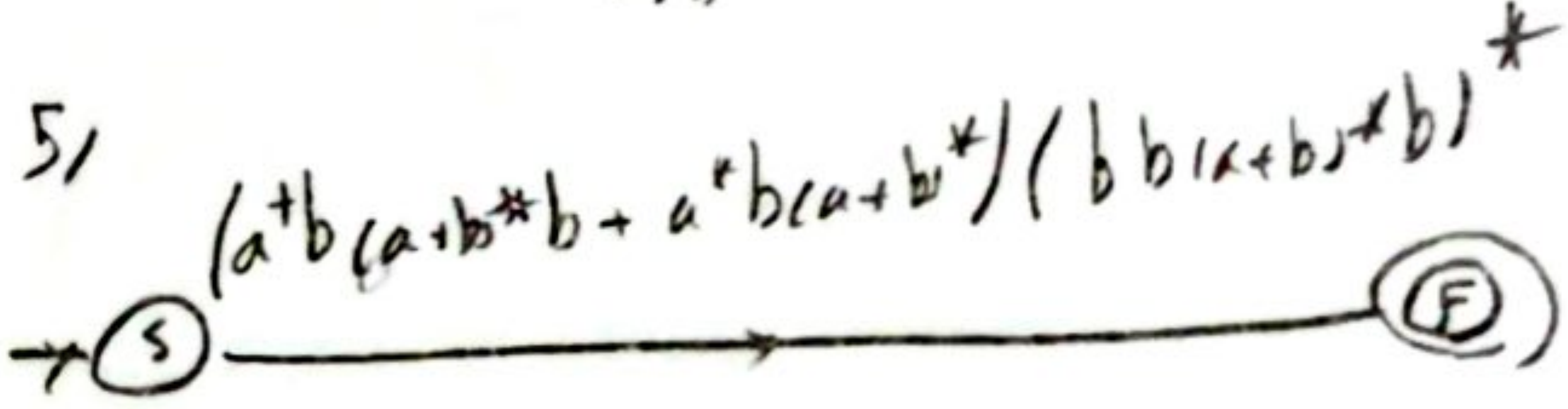
3/



4/

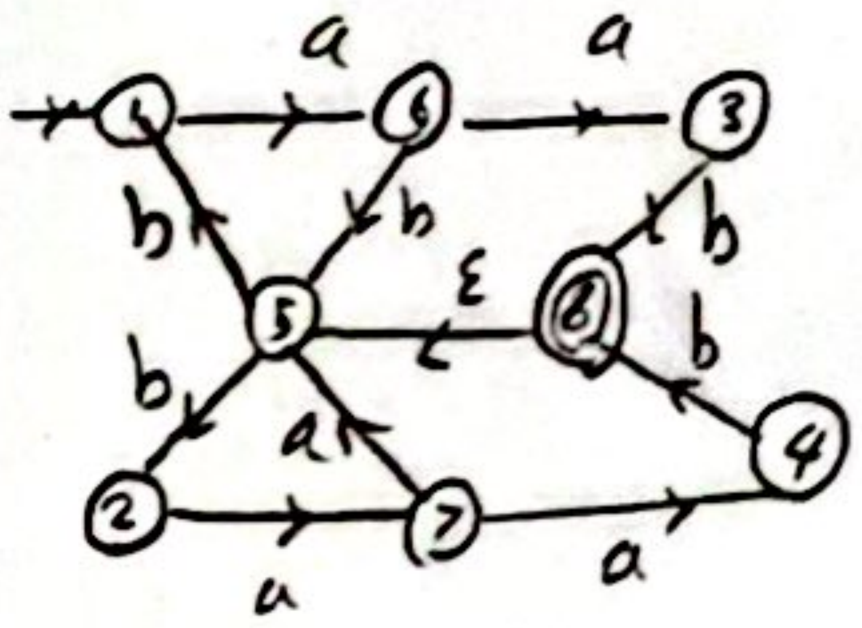


5/

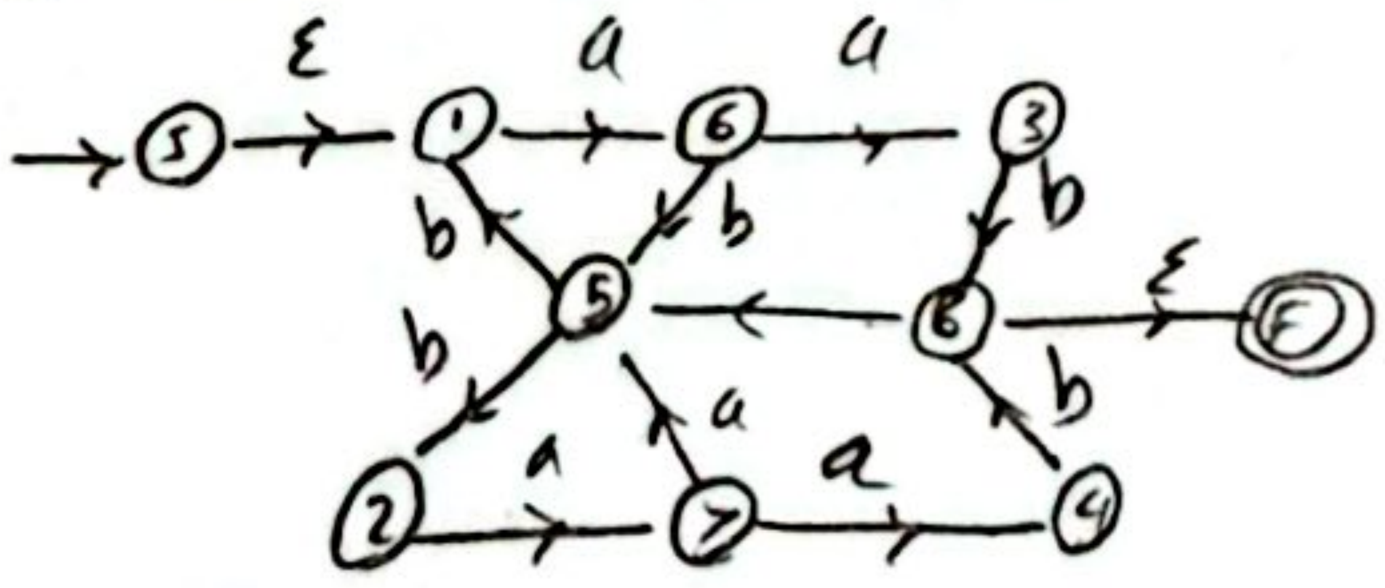


$$R_1 = (a^+b(a+b)^*b + a^*b(a+b)^*)(bb(a+b)^*b)^*$$

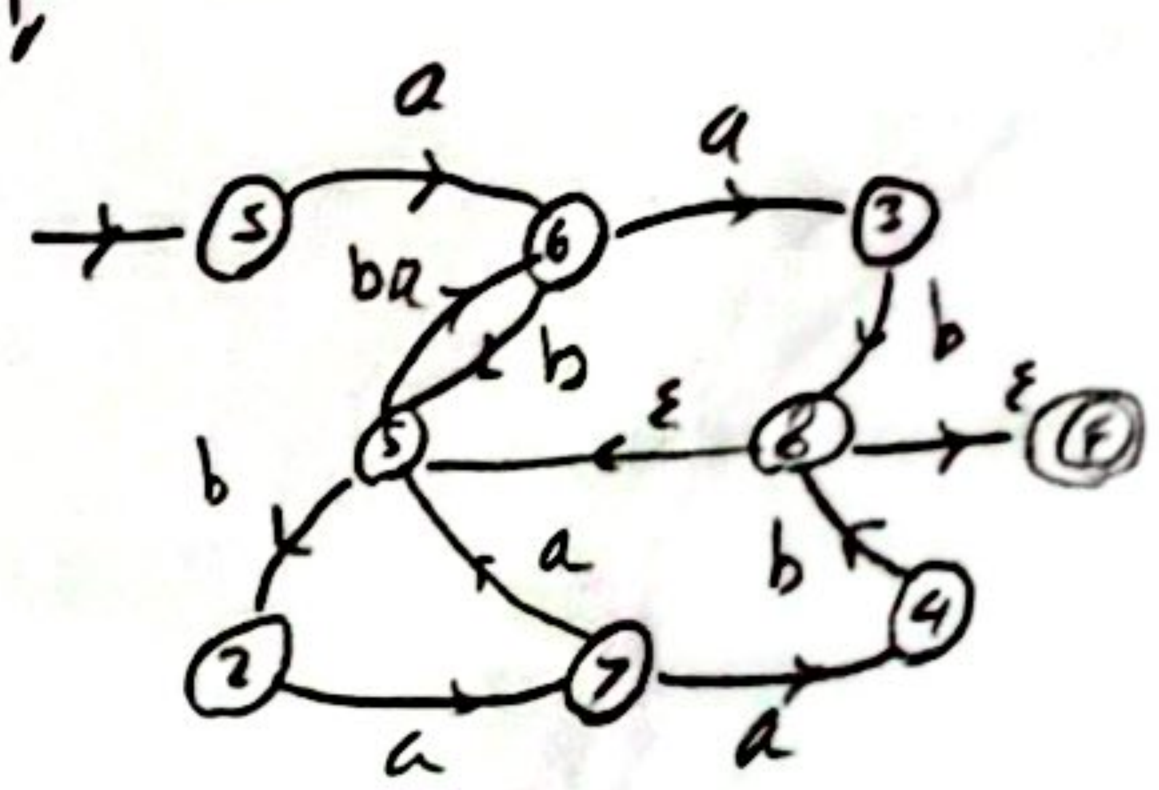
ب) 0/



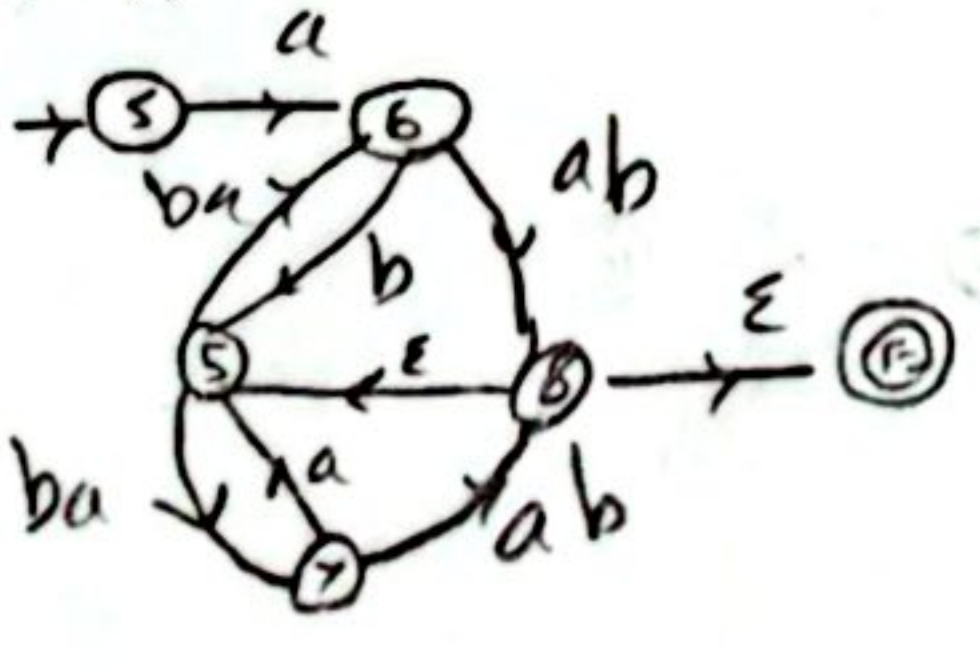
SF/



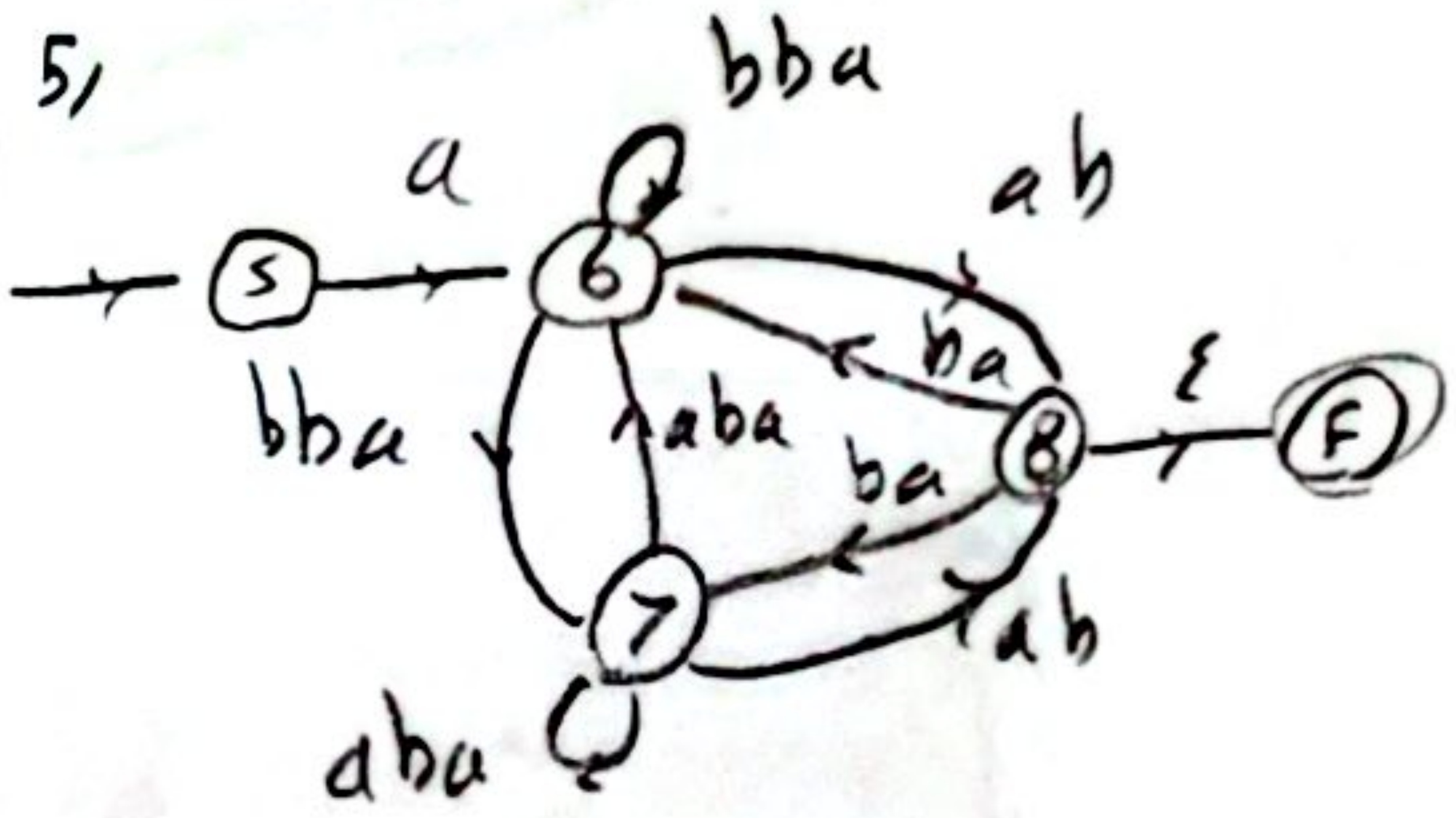
1/



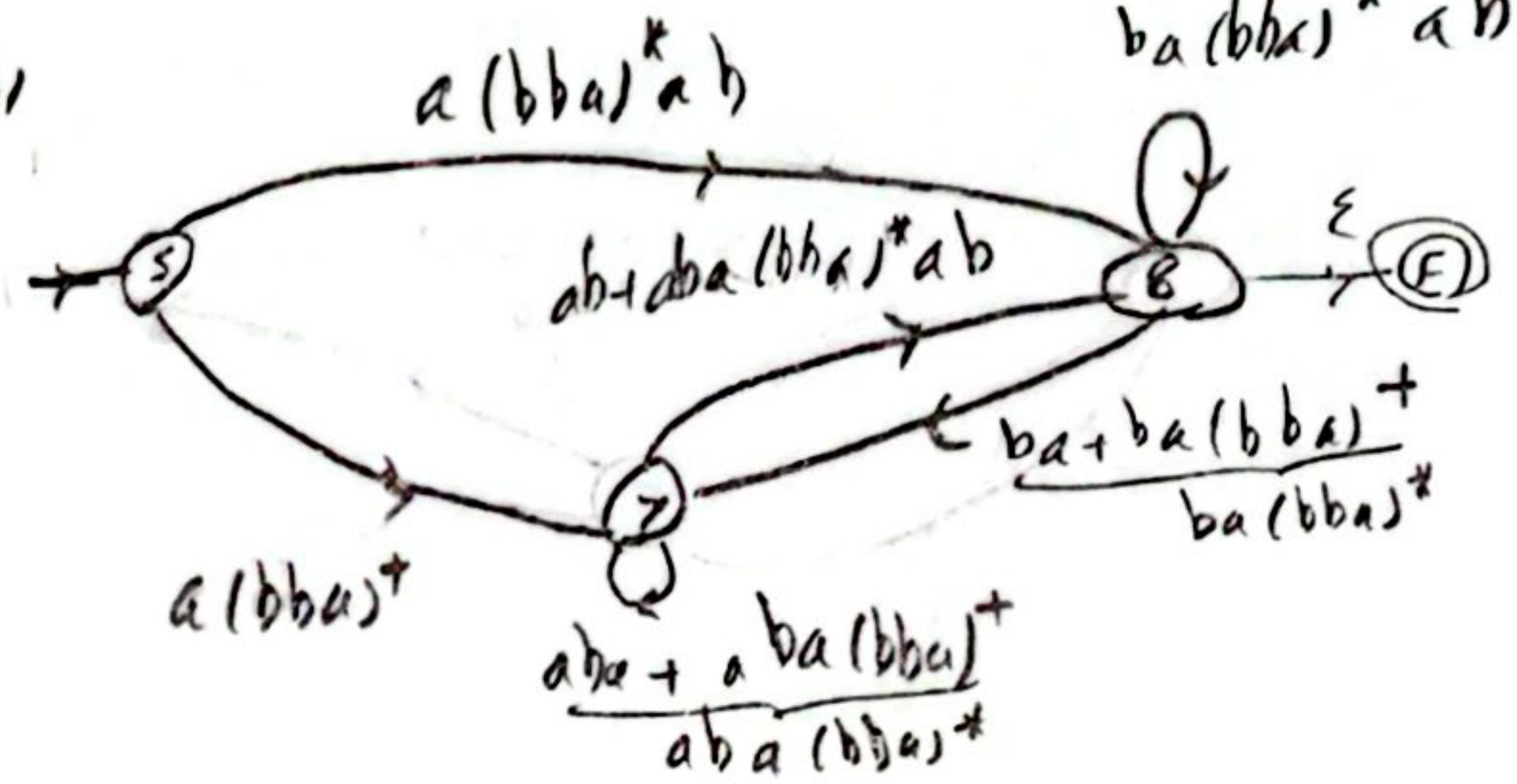
2,3,4/



5/

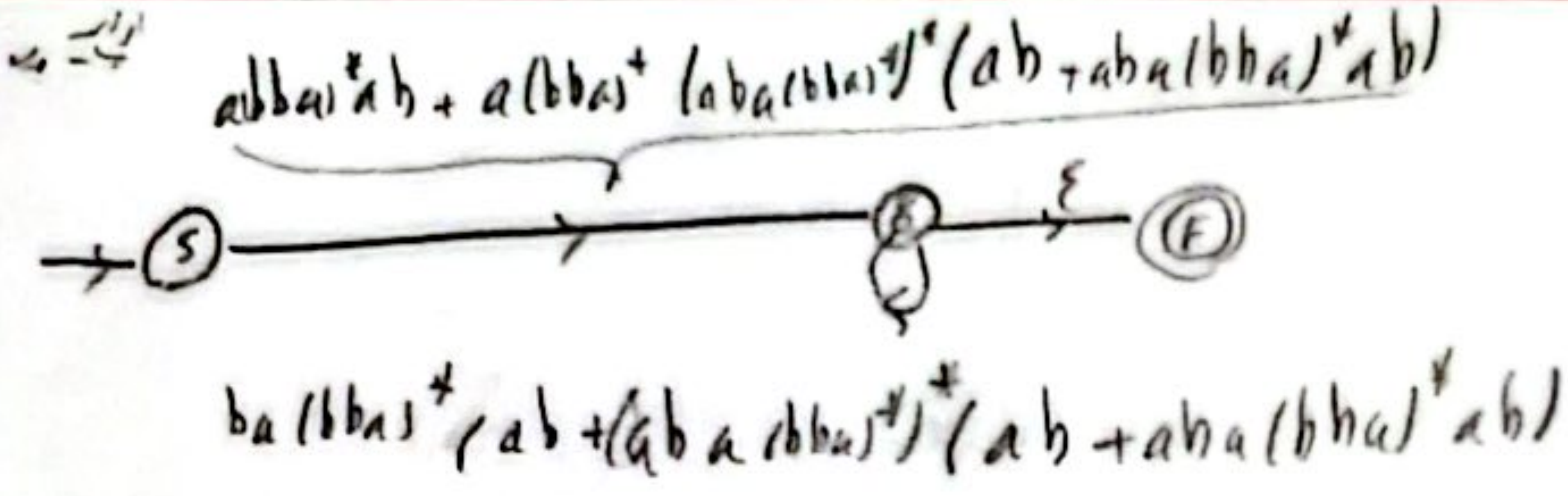


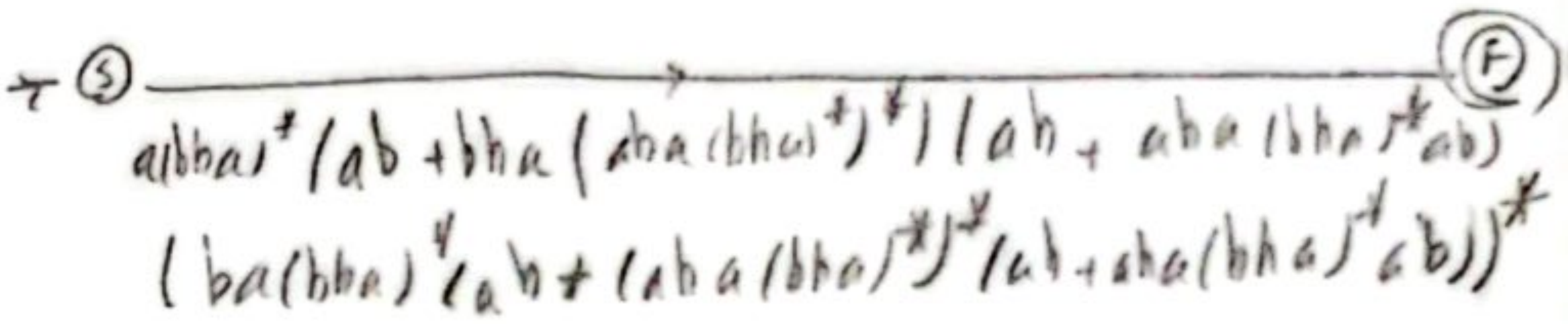
6/


S
$a(bba)^*ab$
$ab+aba(bba)^*ab$
$ba(bba)^*ab$
8
ε
F
$\frac{ba+ba(bba)^+}{ba(bba)^*}$
7
$a(bba)^+$
$\frac{aba+aba(bba)^+}{aba(bba)^*}$

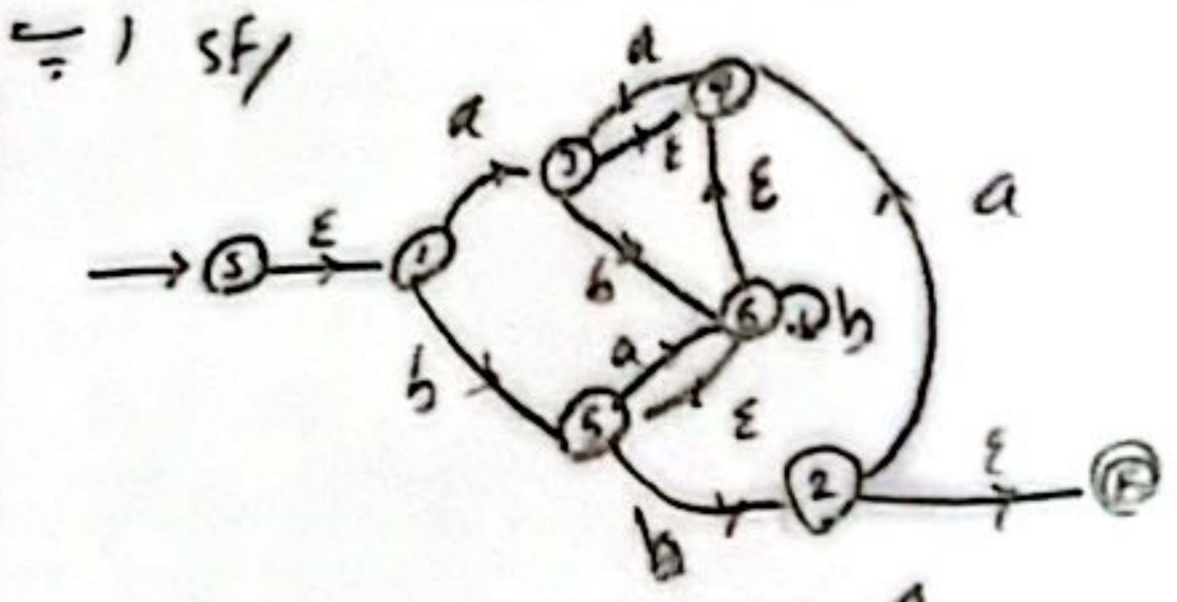
5. ب

7/

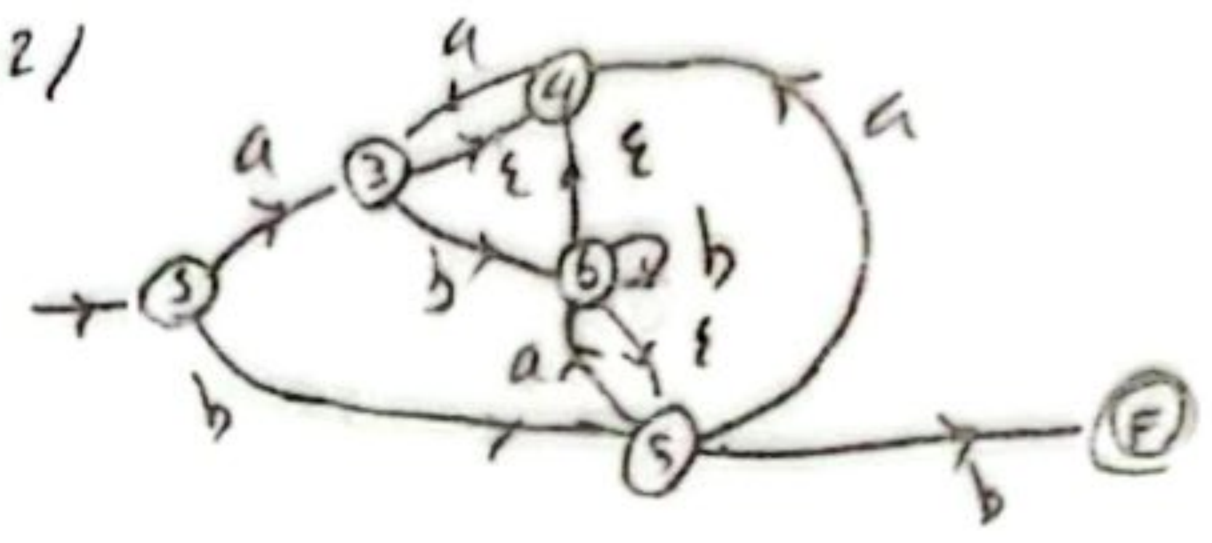

$a(bba)^*ab + a(bba)^+(aba(bba)^*)^*(ab+aba(bba)^*ab)$
S
6
ε
F
$ba(bba)^*(ab+(aba(bba)^*)^*(ab+aba(bba)^*ab)$


S
F
$a(bba)^*(ab+bba(aba(bba)^*)^*)(ab+aba(bba)^*ab)$
$(ba(bba)^*(ab+(aba(bba)^*)^*(ab+aba(bba)^*ab))^*$

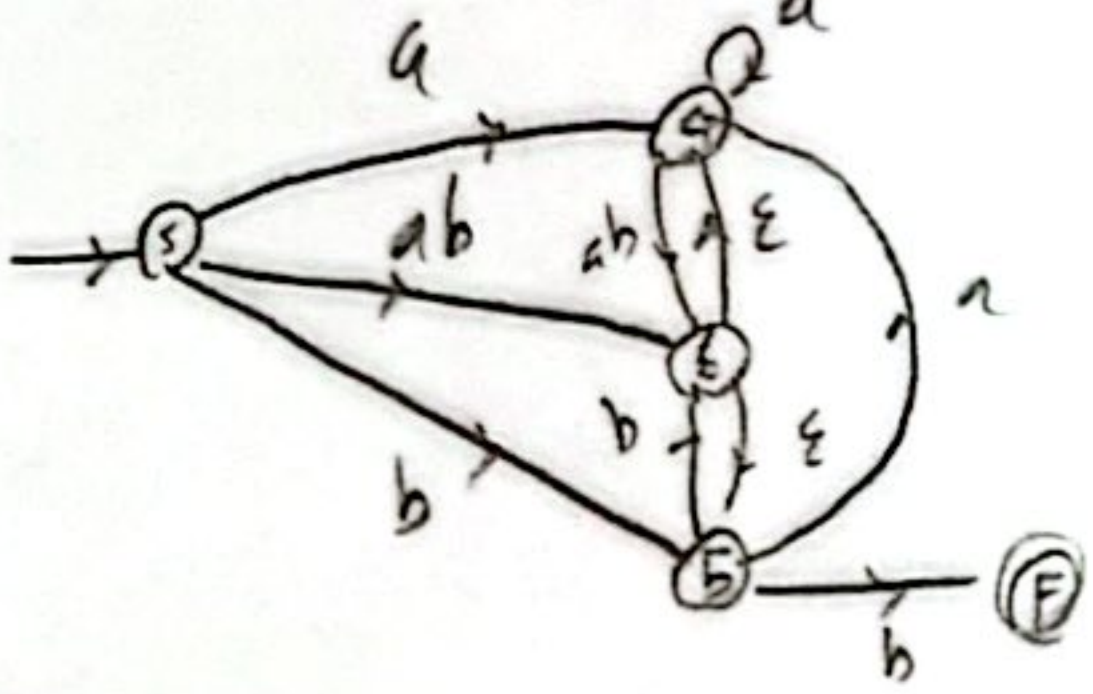
پ) SF/



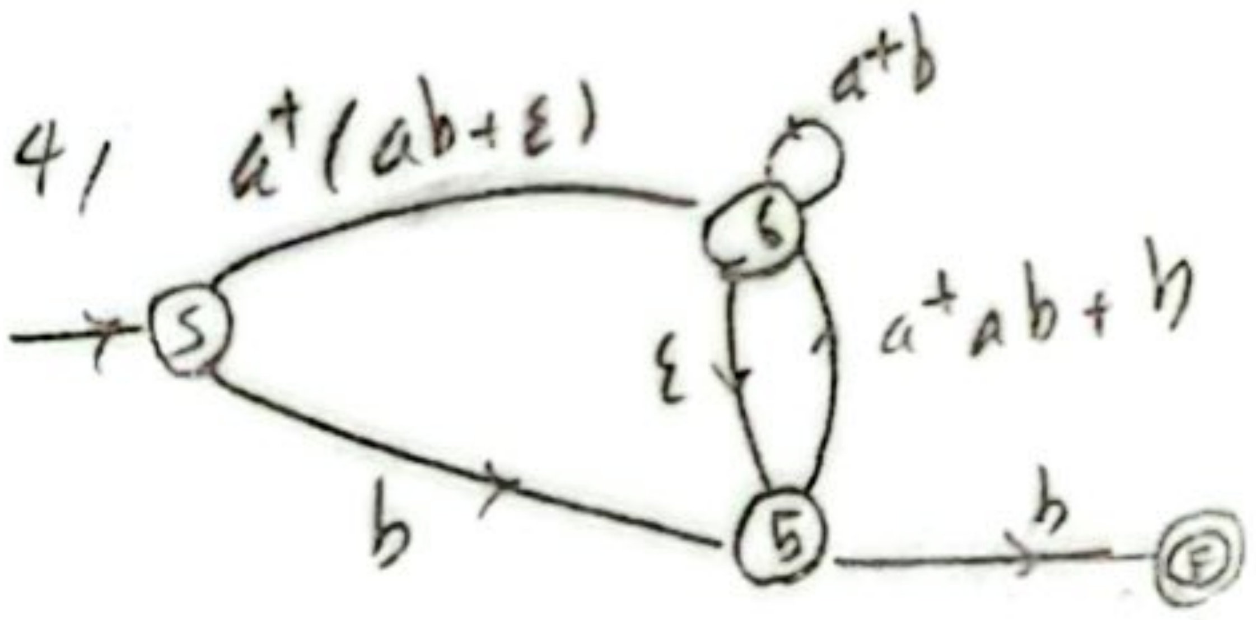
1,2/



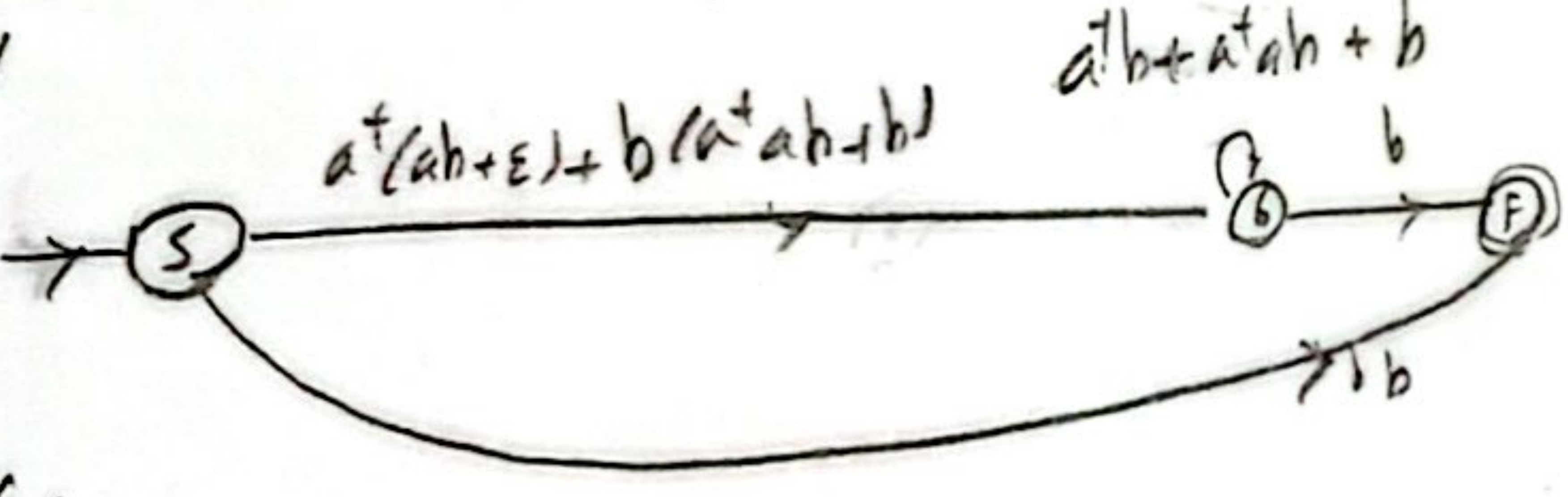
3/



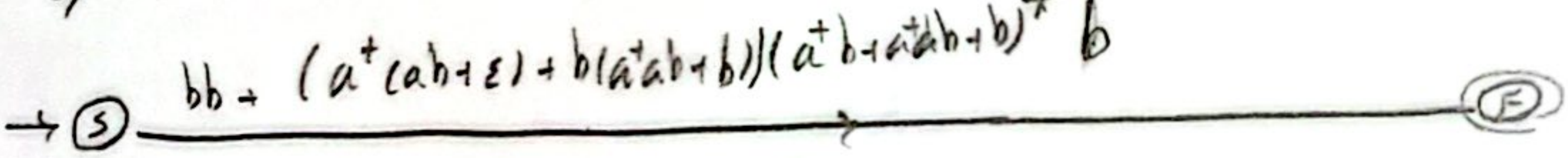
4/



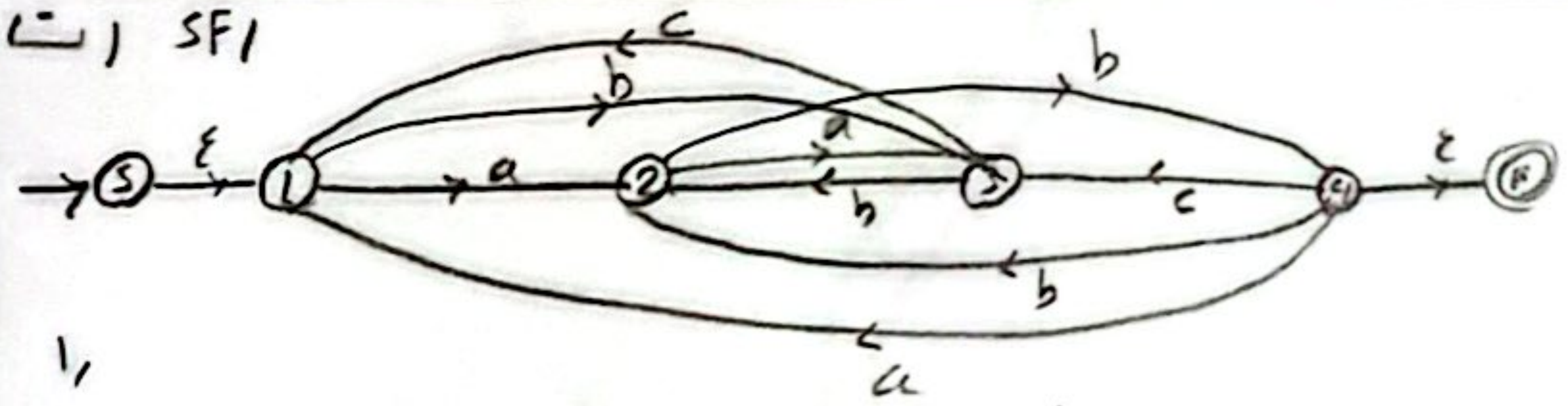
5/



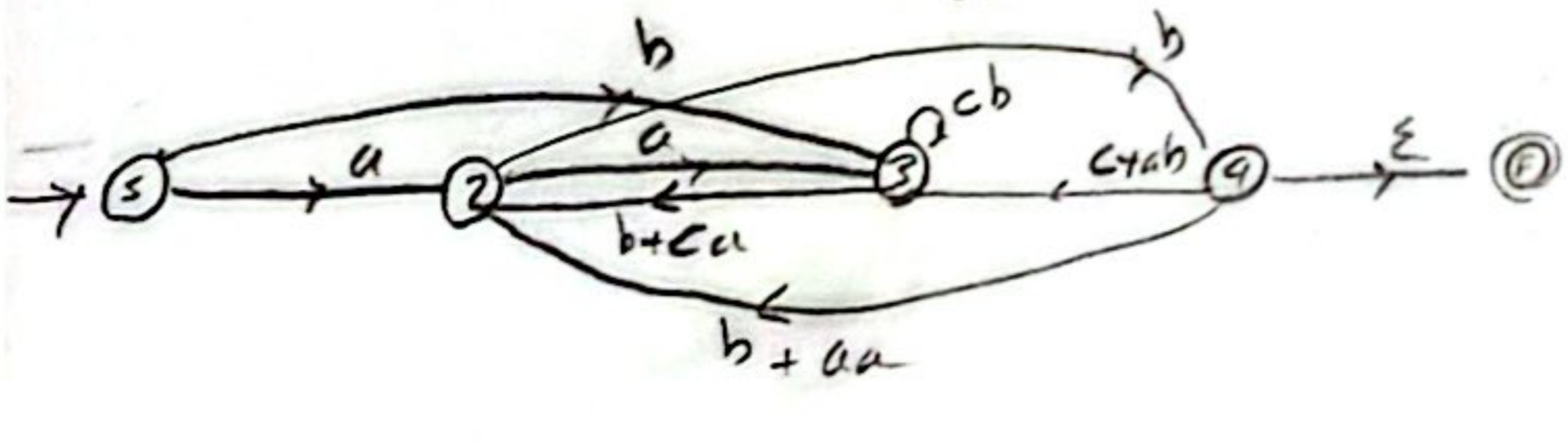
6/



ت) SF/



1/



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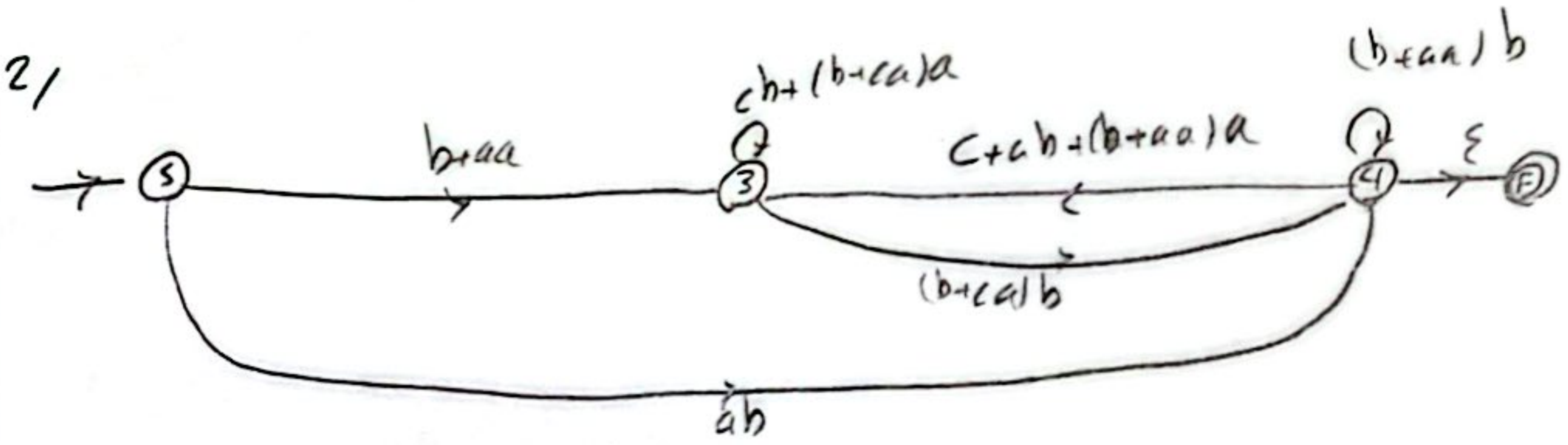
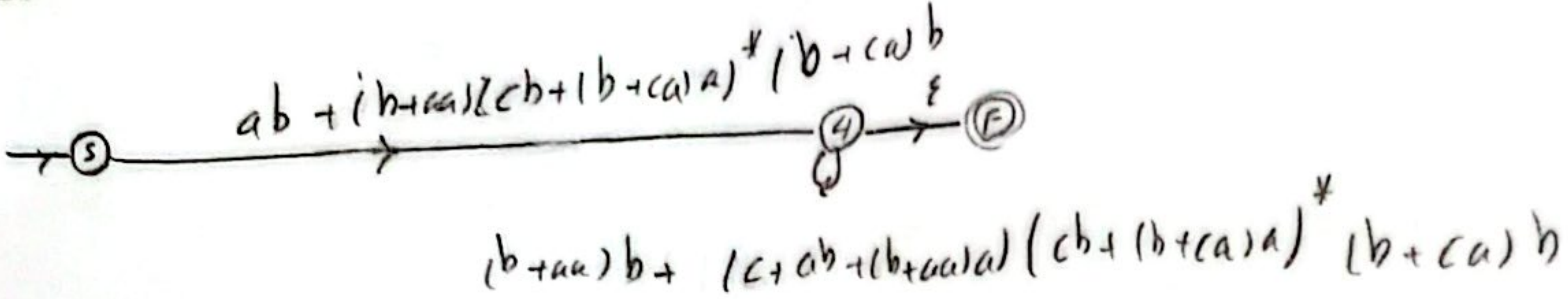
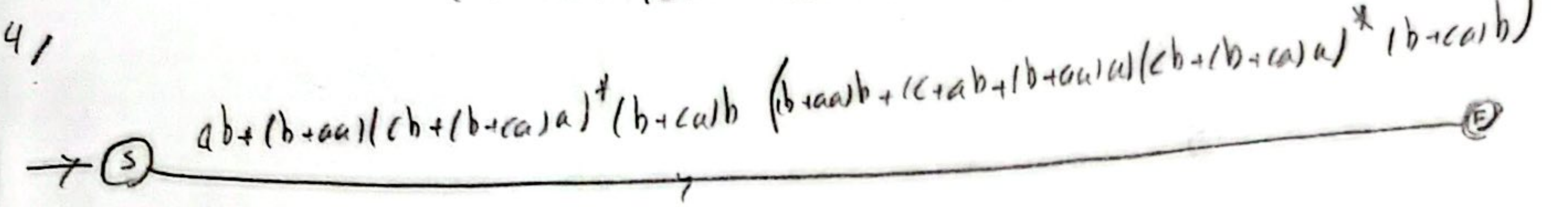


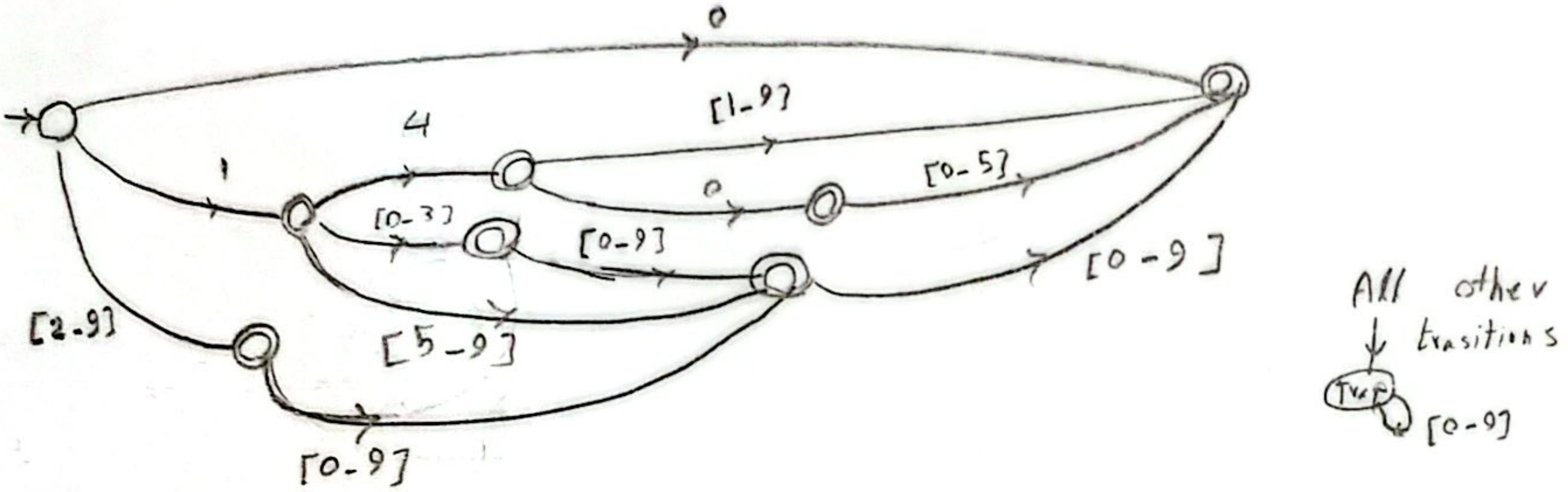
5. الف
3)


ab+(b+aa)(cb+(b+ca)a)*(b+ca)b
ε
S
4
F
(b+aa)b+(c+ab+(b+aa)a)(cb+(b+ca)a)*(b+ca)b

4)


ab+(b+aa)(cb+(b+ca)a)*(b+ca)b ((b+aa)b+(c+ab+(b+aa)a)(cb+(b+ca)a)*(b+ca)b)*
S
F

6)
الف)



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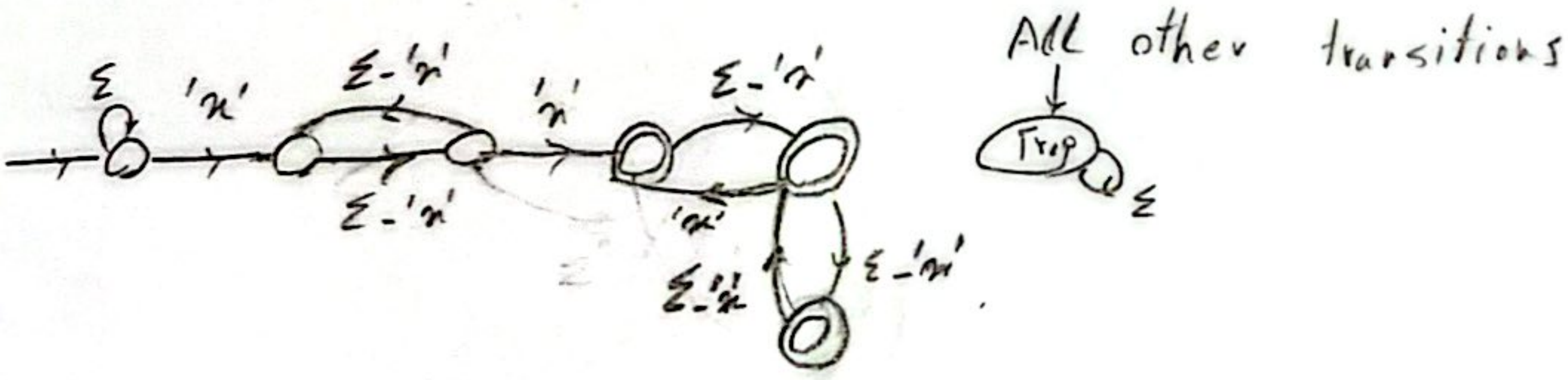

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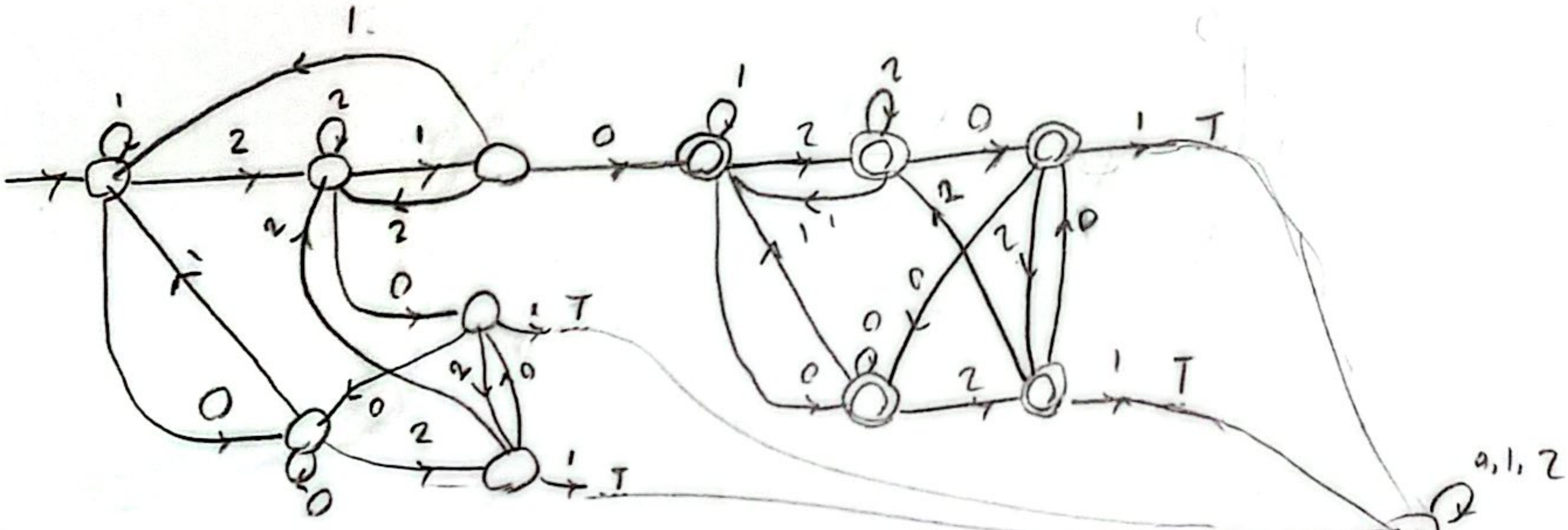

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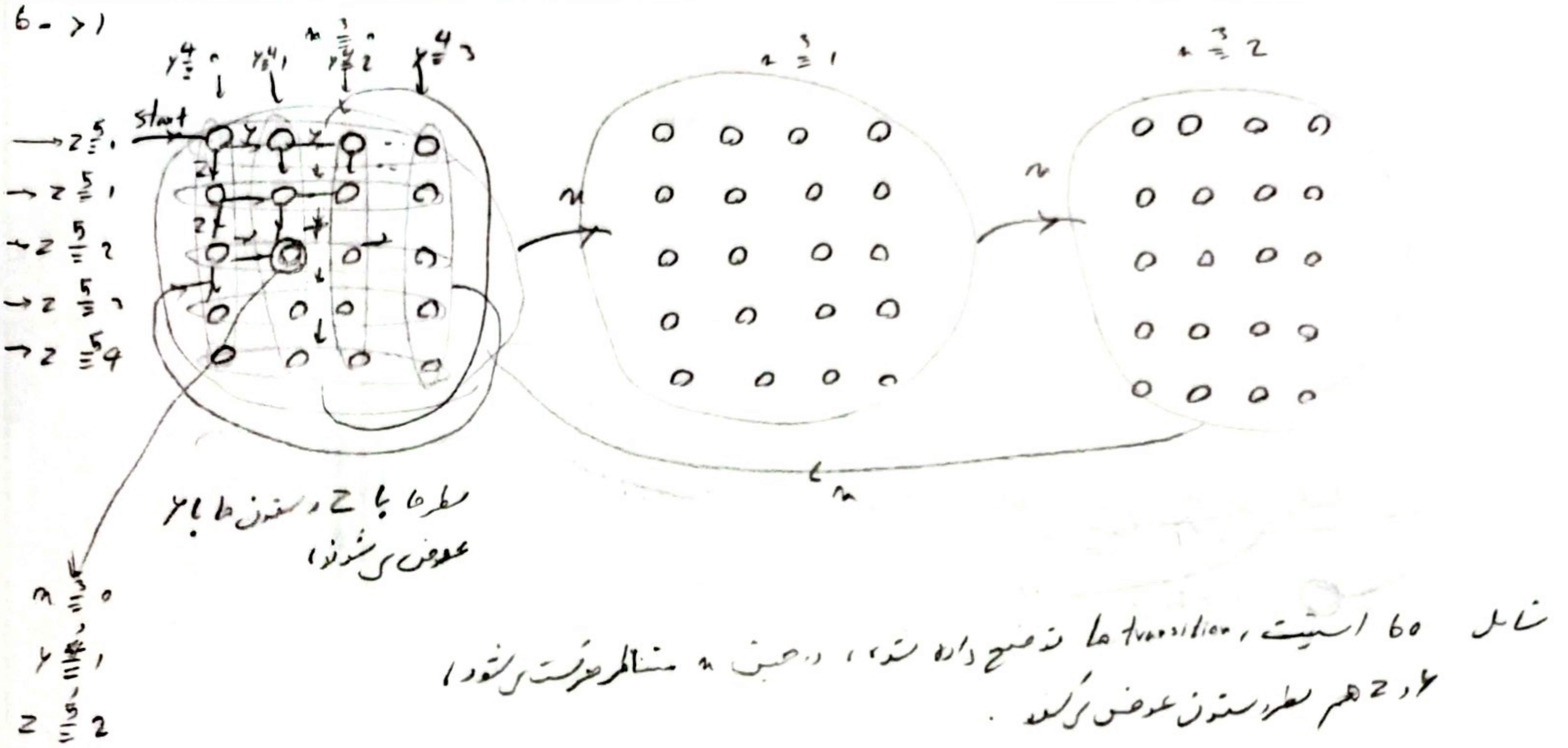
6-۱)



سطرها با z و ستون‌ها با y
عوض می‌شوند

شامل 60 استیت، transition ها توضیح داده شده، در جهت x متناظر حرکت می‌شود،
y و z هم سطر و ستون عوض می‌کنند.

---

۷)

الف) S → RC

```
RC → RC ';' RC | reactiveclass  ID  '(' NUM ')'  '{'  KR  SV  MP '}'
KR → knownrebecs '{'  BC  '}'
BC → BC ';' BC | "Bridge Controller"  ID  ';'
SV → statevars  '{'  VARS '}'
VARS → VARS ';' VARS | TYPE  ID ;
MP → msgsrv ID '(' ')' '{' stmt '}'
Stmt → stmt ';' stmt | if '(' Expr ')' stmt (else Stmt))? | ID '.' ID '(' ID ')'
```

Identifier
number
like int, string, ... → can be ID for simplification

ب) MP → msgsrv ID '(' ')' '{' stmt '}' ⇒ msgsrv passed () { stmt }

```
⇒ msgsrv passed() { if '(' Expr ')' Stmt else Stmt }
⇒ msgsrv passed() { if (id>0) stmt else Stmt }
⇒ msgsrv passed() { if (id>0) if('Stmt') stmt else stmt }
⇒ msgsrv passed() { if (id>0) if(id==1) Stmt else stmt }
⇒ msgsrv passed() { if (id>0) if(id==1) ID.ID(ID); else stmt }
⇒ ______________________________ bc.ID(ID); ______
⇒ ______________________________ bc.arrive(ID); ______
⇒ ______________________________ bc.arrive(id); ______
⇒ ______________________________ else ID.ID(ID); }
⇒ ______________________________ bc.ID(ID); }
⇒ ______________________________ bc.arrive(ID); }
⇒ msgsrv passed() { if (id>0) if(id==1) bc.arrive(id); else bc.arrive(0); }
```

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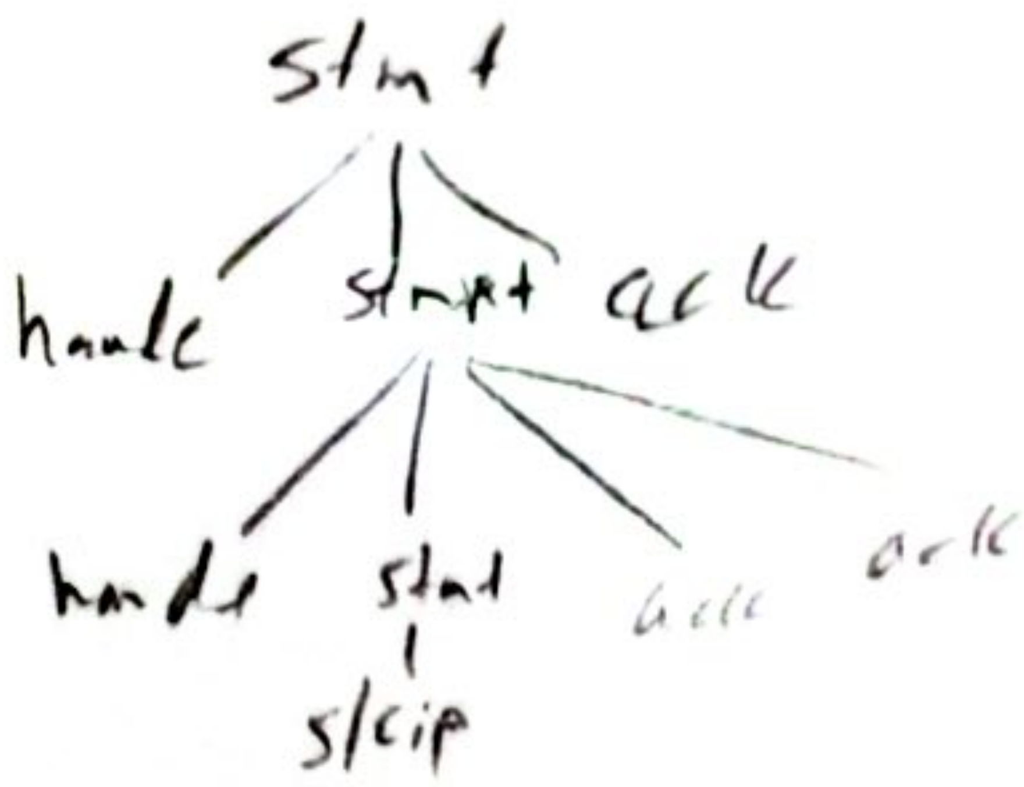
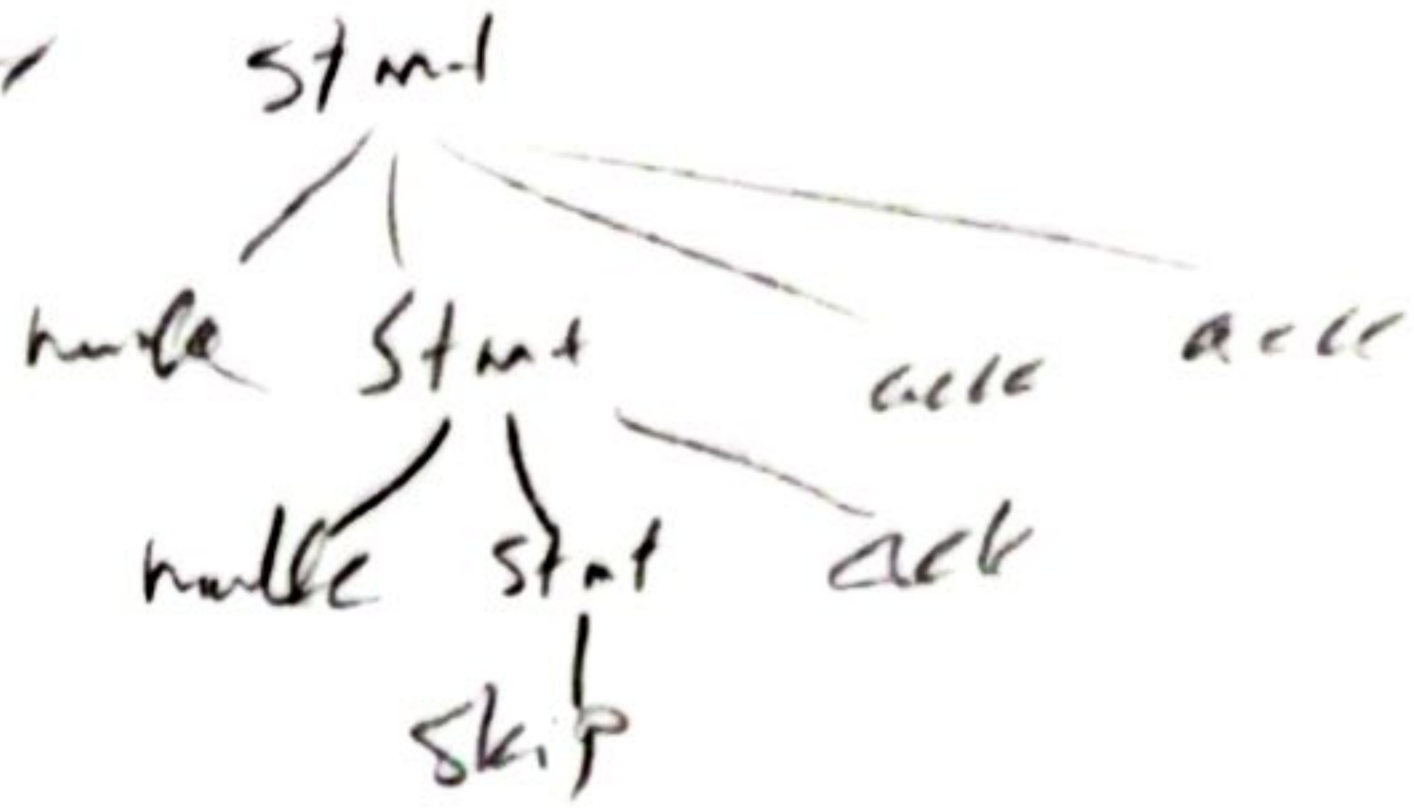
7- i:) handle handle skip ack ack ack

1,



2,



ii) Stmt → handle stmt ack | Double
Double → handle Double ack ack | skip

→ Forces CFG to first take singer ack's and then double ack's